

SAT Math Mastery · Week 5B**Ratios, Rates, Units & Density — Answer Key & Worked Solutions**

Time	120 minutes: 90 min solving + 30 min self-check and error log
Topics	unit conversions (including squared and cubed units) · rates and rate expressions · density · ratios and proportions · scale drawings and scale factors · dimensional analysis

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Convert units with a chain of conversion factors so that unwanted units cancel.
- S2.** Convert squared and cubed units: use the length factor squared for area and cubed for volume.
- S3.** Convert compound rates (distance per time, volume per time, anything per time²).
- S4.** Set up and solve proportions; split a total in a given ratio; compare unit rates.
- S5.** Scale drawings and maps: lengths scale by k , areas by k^2 , volumes by k^3 .
- S6.** Density: $\rho = \frac{m}{V}$; find mass, volume or edge length; combine materials by mass or by volume.
- S7.** Build symbolic rate expressions (“at this rate, how many...”) and test them with numbers.
- S8.** Hidden conditions: relative motion, combined work rates, mixtures, similar solids.

2 · REQUIRED FACTS

- A conversion factor equals 1 (for example $\frac{100 \text{ cm}}{1 \text{ m}}$); multiply so the unit you want to remove sits in the opposite position and cancels.
- If lengths convert by the factor c , then areas convert by c^2 and volumes by c^3 . Rates over time² use the time factor squared.
- 1 L = 1000 cm³, 1 m³ = 1000 L, 1 m³ = 10⁶ cm³, 1 h = 3600 s, 1 km/h = $\frac{1}{3.6}$ m/s.
- Scale 1 : k (or “1 unit represents k units”): actual length = $k \times$ drawn length; actual area = $k^2 \times$ drawn area.
- Densities do not average: density of a mixture is $\frac{\text{total mass}}{\text{total volume}}$.
- Combined work rates add: if A needs a hours and B needs b hours, together they do $\frac{1}{a} + \frac{1}{b}$ of the job per hour.

3 · REQUIRED FORMULAS

$$\begin{aligned} \text{rate} &= \frac{\text{quantity}}{\text{time}}, & d &= rt, & \rho &= \frac{m}{V}, & m &= \rho V, & V_{\text{cube}} &= s^3, & A_{\text{cube}} &= 6s^2 \\ \frac{\text{part}}{\text{whole}} &= \frac{a}{a+b} \text{ for ratio } a:b, & \frac{A_2}{A_1} &= k^2, & \frac{V_2}{V_1} &= k^3, & \frac{m_2}{m_1} &= k^3 \text{ (same material)} \\ V_{\text{cylinder}} &= \pi r^2 h, & \text{flow rate} &= \text{cross-section area} \times \text{speed}, & \rho_{\text{mix}} &= \frac{m_1 + m_2}{V_1 + V_2} \end{aligned}$$

4 · MINI THEORY SHEET

Unit chains. Write the starting quantity, then multiply by fractions that equal 1, each cancelling one unit.

Example: $72 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 20 \frac{\text{m}}{\text{s}}$.

Squared and cubed units. 1 m = 100 cm $\Rightarrow$ 1 m² = 100² cm² = 10⁴ cm² and 1 m³ = 10⁶ cm³. Converting 0.6 km² to m² multiplies by 1000², not 1000.

Scale. Drawing-to-actual factor k for length gives k^2 for area and k^3 for volume or mass. To go between two maps, convert through the actual size, or multiply the two length factors and square.

Ratios and rates. A ratio $a : b$ splits a total T into $\frac{a}{a+b}T$ and $\frac{b}{a+b}T$. For “at this rate” questions, find the unit rate first, then scale.

Density. Mass = density $\times$ volume. To find an edge, find the volume first and then take the cube root. Mixtures: add masses and add volumes, then divide.

Symbolic rates. Write the rate with its units (boxes per minute), convert the time to match, multiply. Verify with simple numbers.

Speed shortcuts.

Situation	Shortcut
area / volume units	multiply by the length factor ² / length factor ³
km/h $\leftrightarrow$ m/s	divide by 3.6 / multiply by 3.6
rate over time ²	the time factor is squared: $\text{m/s}^2 \rightarrow \text{km/h}^2$ multiplies by $\frac{1}{1000} \cdot 3600^2$
scale factor from area	$k = \sqrt{\text{area ratio}}$; then mass or volume ratio is k^3
ratio $a : b$ of a total	fraction $\frac{a}{a+b}$ of the total; do not divide by b
catch-up problems	head start = (speed difference) $\times$ time
symbolic rate check	substitute easy numbers and test each option
mixtures	$\rho_{\text{mix}} = \frac{\text{total mass}}{\text{total volume}}$, never the average of densities

5 · GUIDED EXAMPLES

Example 1 — Cubed units

A tank holds 2.4 m^3 of water. How many liters is that? ($1 \text{ m} = 100 \text{ cm}$, $1 \text{ L} = 1000 \text{ cm}^3$)

$$2.4 \text{ m}^3 = 2.4 \times 100^3 \text{ cm}^3 = 2.4 \times 10^6 \text{ cm}^3. \text{ Then } \frac{2.4 \times 10^6}{1000} = 2400 \text{ L.}$$

Example 2 — Scale and area

On a map, 1 cm represents 5 km . A lake covers 6 cm^2 on the map. Find its actual area.

Lengths scale by 5, so areas scale by $5^2 = 25$: $6 \times 25 = 150 \text{ km}^2$ (not $6 \times 5 = 30$).

Example 3 — Density of a mixture

A 60 g piece of metal with density 12 g/cm^3 is melted with a 30 g piece with density 3 g/cm^3 . Find the density of the mixture (volumes add).

$$\text{Volumes: } \frac{60}{12} + \frac{30}{3} = 5 + 10 = 15 \text{ cm}^3. \text{ Mass: } 90 \text{ g. Density: } \frac{90}{15} = 6 \text{ g/cm}^3, \text{ not the average } 7.5.$$

6 · PRACTICE SET — 30 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · CALCULATOR

A car travels at a constant speed and covers 270 miles in 4.5 hours. At this rate, how many miles will the car travel in 7 hours?

Worked solution

1. Unit rate: $\frac{270}{4.5} = 60$ miles per hour.

2. $60 \times 7 = 420$ miles.

Answer: 420

2. EASY · CALCULATOR

A board is 30 inches long. Given that 1 inch = 2.54 centimeters, what is the length of the board, in centimeters?

(A) 32.54

(B) 76.2 ✓

(C) 762

(D) 11.81

Worked solution

1. Multiply by the factor $\frac{2.54 \text{ cm}}{1 \text{ in}}$ so that inches cancel.

2. $30 \times 2.54 = 76.2$ cm.

Answer: (B)

3. EASY · CALCULATOR

A metal block has a volume of 12 cm^3 and a mass of 94.8 grams. What is the density of the metal, in grams per cubic centimeter?

(A) 1137.6

(B) 0.127

(C) 7.9 ✓

(D) 82.8

Worked solution

1. $\rho = \frac{m}{V} = \frac{94.8}{12}$.

2. $\rho = 7.9 \text{ g/cm}^3$.

Answer: (C)

4. EASY · NO CALCULATOR NEEDED

The table shows map distances and the corresponding actual distances for a map that has a constant scale. What actual distance, in kilometers, corresponds to a map distance of 7 centimeters?

Map distance (cm)	2	5	8
Actual distance (km)	30	75	120

Worked solution

1. Scale: $\frac{30}{2} = \frac{75}{5} = \frac{120}{8} = 15$ km per cm.

2. $7 \times 15 = 105$ km.

Answer: 105

5. EASY · NO CALCULATOR NEEDED
A conveyor belt moves at a constant speed of 0.8 meters per second. At this speed, how many meters does the belt move in 1 hour?

- (A) 2880 ✓ (B) $\frac{1}{4500}$
(C) 48 (D) 288

Worked solution

1. 1 hour = $60 \cdot 60 = 3600$ seconds.

2. $0.8 \times 3600 = 2880$ meters (not $0.8 \times 60 = 48$, which is one minute).

Answer: (A)

6. MEDIUM · NO CALCULATOR NEEDED
A rectangular rug has an area of 2.5 square feet. Given that 1 foot = 12 inches, what is the area of the rug, in square inches?

- (A) 4320 (B) $\frac{2.5}{144}$
(C) 30 (D) 360 ✓

Worked solution

1. Areas convert with the factor squared: $1 \text{ ft}^2 = 12^2 = 144 \text{ in}^2$.

2. $2.5 \times 144 = 360$ square inches.

Answer: (D)

7. MEDIUM · NO CALCULATOR NEEDED
A storage tank has a volume of 3.5 cubic meters. How many liters of water does the tank hold when full? (1 meter = 100 centimeters and 1 liter = 1000 cubic centimeters.)

Worked solution

1. $1 \text{ m}^3 = 100^3 \text{ cm}^3 = 10^6 \text{ cm}^3$, so $3.5 \text{ m}^3 = 3.5 \times 10^6 \text{ cm}^3$.

2. Divide by 1000 cm^3 per liter: $\frac{3.5 \times 10^6}{1000} = 3500$ liters.

Answer: 3500

8. MEDIUM · NO CALCULATOR NEEDED
A train travels at a constant speed of 72 kilometers per hour. What is the speed of the train, in meters per second?

- (A) 259.2 (B) 1.2
(C) 20 ✓ (D) 72

Worked solution

1. $72 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}}$.

2. $= \frac{72,000}{3600} = 20$ meters per second (shortcut: divide by 3.6).

Answer: (C)

9.

MEDIUM · NO CALCULATOR NEEDED

On a scale drawing of a building, 0.75 inch represents 6 feet. A wall measures 4.5 inches on the drawing. What is the actual length of the wall, in feet?

(A) 27

(B) 36 ✓

(C) $\frac{9}{16}$

(D) 6

Worked solution

1. Feet per inch: $\frac{6}{0.75} = 8$.

2. $4.5 \times 8 = 36$ feet.

Answer: (B)

10.

MEDIUM · CALCULATOR

Aluminum has a density of 2.7 grams per cubic centimeter. What is the mass, in grams, of a solid aluminum cube with edge length 5 centimeters?

(A) 13.5

(B) 40.5

(C) 67.5

(D) 337.5 ✓

Worked solution

1. Volume: $5^3 = 125 \text{ cm}^3$.

2. $m = \rho V = 2.7 \times 125 = 337.5$ grams.

Answer: (D)

11.

MEDIUM · CALCULATOR

Flour and sugar are mixed in the ratio 5 to 3 by mass. A batch of the mixture has a total mass of 4.8 kilograms. How many kilograms of sugar are in the batch?

Worked solution

1. Sugar is $\frac{3}{5+3} = \frac{3}{8}$ of the total.

2. $\frac{3}{8} \times 4.8 = 1.8$ kilograms.

Answer: 1.8

12.

MEDIUM · NO CALCULATOR NEEDED

A machine fills b boxes every m minutes. At this rate, which of the following expressions gives the number of boxes the machine fills in h hours?

(A) $\frac{60bh}{m}$ ✓(B) $\frac{60hm}{b}$ (C) $\frac{bm}{60h}$ (D) $\frac{bh}{60m}$ **Worked solution**

1. Rate: $\frac{b}{m}$ boxes per minute. h hours = $60h$ minutes.

2. Boxes = $\frac{b}{m} \cdot 60h = \frac{60bh}{m}$.

3. Check with $b = 12$, $m = 5$, $h = 2$: $\frac{12}{5} \cdot 120 = 288$ and $\frac{60(12)(2)}{5} = 288$.

Answer: (A)

13.

MEDIUM · NO CALCULATOR NEEDED

Pump A can fill a tank in 8 hours working alone. Pump B can fill the same tank in 24 hours working alone. How many hours will it take the two pumps, working together, to fill the tank?

Worked solution

1. In one hour: A fills $\frac{1}{8}$ and B fills $\frac{1}{24}$ of the tank.

2. Together: $\frac{1}{8} + \frac{1}{24} = \frac{3}{24} + \frac{1}{24} = \frac{1}{6}$ of the tank per hour.

3. Time = 6 hours.

Answer: 6

14.

HARD · NO CALCULATOR NEEDED

Water leaks from a reservoir at a constant rate of 2.4 cubic meters per hour. At this rate, how many liters of water leak out per minute? (1 cubic meter = 1000 liters.)

(A) 4

(B) 40 ✓

(C) 144

(D) 2400

Worked solution

1. $2.4 \frac{\text{m}^3}{\text{h}} \times \frac{1000 \text{ L}}{1 \text{ m}^3} = 2400$ liters per hour.

2. Divide by 60 minutes per hour: $\frac{2400}{60} = 40$ liters per minute.

Answer: (B)

15.

HARD · CALCULATOR

Solar panels, each covering 2 square meters, are installed so that they cover 40% of a solar farm. The farm has an area of 0.6 square kilometers. How many panels are installed? (1 kilometer = 1000 meters.)

Worked solution

1. $0.6 \text{ km}^2 = 0.6 \times 1000^2 = 600,000 \text{ m}^2$ (square the factor).

2. Covered area: $0.4 \times 600,000 = 240,000 \text{ m}^2$.

3. Panels: $\frac{240,000}{2} = 120,000$.

Answer: 120000

16.

HARD · CALCULATOR

A solid lead cube has a mass of 2440.8 grams. The density of lead is 11.3 grams per cubic centimeter. What is the edge length of the cube, in centimeters?

(A) 72

(B) 216

(C) 6 ✓

(D) 36

Worked solution

1. Volume: $V = \frac{m}{\rho} = \frac{2440.8}{11.3} = 216 \text{ cm}^3$.

2. Edge: $s = \sqrt[3]{216} = 6 \text{ cm}$. (Not 216, the volume, and not $\sqrt{216}$.)

Answer: (C)

17.

HARD · CALCULATOR

A model of a bridge is built at a scale of 1 : 40. The total surface area of the model is 150 square centimeters. What is the surface area of the actual bridge, in square meters?

(A) 24 ✓

(B) 2400

(C) 240,000

(D) 0.6

Worked solution

1. Lengths scale by 40, so areas scale by $40^2 = 1600$.

2. Actual area: $150 \times 1600 = 240,000 \text{ cm}^2$.

3. Convert with $1 \text{ m}^2 = 100^2 = 10,000 \text{ cm}^2$: $\frac{240,000}{10,000} = 24 \text{ m}^2$.

Answer: (A)

18.

HARD · CALCULATOR

Vehicle A travels at 54 kilometers per hour, vehicle B at 16 meters per second, and vehicle C at 0.93 kilometers per minute. Which of the following lists the vehicles in order from slowest to fastest?

(A) B, C, A

(B) A, B, C

(C) C, A, B

(D) A, C, B ✓

Worked solution

1. Convert everything to meters per second.

2. A: $\frac{54,000}{3600} = 15$. B: 16. C: $\frac{930}{60} = 15.5$.

3. Slowest to fastest: A (15), C (15.5), B (16).

Answer: (D)

19.

HARD · NO CALCULATOR NEEDED

A mixture is made of concentrate and water in the ratio 2 to 7 by volume, and 18 liters of it is prepared. How many liters of concentrate must be added to the 18 liters so that the ratio of concentrate to water becomes 1 to 2?

Worked solution

1. Concentrate: $\frac{2}{9} \cdot 18 = 4 \text{ L}$. Water: $\frac{7}{9} \cdot 18 = 14 \text{ L}$.

2. Adding x liters of concentrate leaves the water at 14 L: $\frac{4+x}{14} = \frac{1}{2}$.

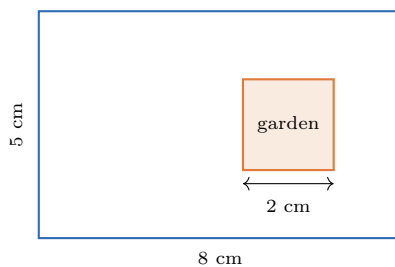
3. $4+x=7 \Rightarrow x=3$ liters.

Answer: 3

20.

HARD · NO CALCULATOR NEEDED

The scale drawing shows a rectangular plot of land that contains a square garden. On the drawing, 1 centimeter represents 3 meters. What is the actual area, in square meters, of the part of the plot that is *not* the garden?



(A) 36

(B) 108

(C) 324 ✓

(D) 972

Worked solution

1. Drawing areas: plot $8 \times 5 = 40 \text{ cm}^2$, garden $2 \times 2 = 4 \text{ cm}^2$; the rest is 36 cm^2 .

2. Lengths scale by 3, so areas scale by $3^2 = 9$.

3. $36 \times 9 = 324 \text{ m}^2$.

Answer: (C)

21.

HARD · CALCULATOR

A car travels 32 miles per gallon of fuel. Use 1 mile = 1.609 kilometers and 1 gallon = 3.785 liters. To the nearest tenth, how many kilometers does the car travel per liter of fuel?

Worked solution

1. $32 \frac{\text{mi}}{\text{gal}} \times \frac{1.609 \text{ km}}{1 \text{ mi}} \times \frac{1 \text{ gal}}{3.785 \text{ L}}$

2. $= \frac{32(1.609)}{3.785} = \frac{51.488}{3.785} \approx 13.60$.

3. To the nearest tenth: 13.6 km per liter.

Answer: 13.6

22.

HARD · NO CALCULATOR NEEDED

The edge length of a cube is tripled. Which of the following statements must be true?

I. The surface area of the cube is multiplied by 9.

II. The volume of the cube is multiplied by 27.

III. The ratio of surface area to volume is multiplied by 3.

(A) I only

(B) I and II only ✓

(C) II and III only

(D) I, II and III

Worked solution

1. Let the edge be s : surface area $6s^2 \rightarrow 6(3s)^2 = 54s^2$, a factor of 9. I is true.

2. Volume $s^3 \rightarrow (3s)^3 = 27s^3$, a factor of 27. II is true.

3. Ratio $\frac{6s^2}{s^3} = \frac{6}{s} \rightarrow \frac{6}{3s}$: it is multiplied by $\frac{1}{3}$, not 3. III is false.

Answer: (B)

23.

HARD · CALCULATOR

A ball's speed increases at a constant rate of 9.8 meters per second each second, that is, an acceleration of 9.8 m/s^2 . What is this acceleration in kilometers per hour squared (km/h^2)?

Worked solution

1. Meters to kilometers: multiply by $\frac{1}{1000}$.
2. The time unit is squared: $1 \text{ s}^{-2} = 3600^2 \text{ h}^{-2}$ (since $1 \text{ s} = \frac{1}{3600} \text{ h}$).
3. $9.8 \times \frac{3600^2}{1000} = 9.8 \times 12,960 = 127,008 \text{ km/h}^2$.

Answer: 127008

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

Metal X has a density of 8.0 g/cm^3 and metal Y has a density of 3.0 g/cm^3 . An alloy is made by combining X and Y in the mass ratio 3 : 2 (X to Y). Assuming the volumes simply add, what is the density of the alloy, in g/cm^3 ?

- (A) 5.0 (B) 5.5
(C) 6.0 (D) 4.8 ✓

Worked solution

1. Take 3 g of X and 2 g of Y (mass ratio 3 : 2).
2. Volumes: $\frac{3}{8} + \frac{2}{3} = \frac{9}{24} + \frac{16}{24} = \frac{25}{24} \text{ cm}^3$.
3. Density = $\frac{5}{25/24} = \frac{24}{5} = 4.8 \text{ g/cm}^3$. (Weighting the densities by mass gives 6.0, which is not the alloy density.)

Answer: (D)

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

On Map P, 1 centimeter represents 2 kilometers. On Map Q, 1 centimeter represents 5 kilometers. On Map P, a park occupies 40% of a square region whose side measures 7.5 centimeters. What is the area of the park on Map Q, in square centimeters?

Worked solution

1. Square on Map P: $7.5^2 = 56.25 \text{ cm}^2$; park: $0.4 \times 56.25 = 22.5 \text{ cm}^2$.
2. Actual area: lengths scale by 2, so $22.5 \times 2^2 = 90 \text{ km}^2$.
3. On Map Q lengths shrink by 5, so areas shrink by 25: $\frac{90}{25} = 3.6 \text{ cm}^2$. (Equivalently, $22.5 \times \left(\frac{2}{5}\right)^2 = 3.6$.)

Answer: 3.6

26.

SAT CHALLENGE · CALCULATOR

A freight train leaves a station traveling at a constant 60 kilometers per hour. 1.5 hours later, a second train leaves the same station on the same track, in the same direction, at a constant 25 meters per second. How many kilometers from the station will the second train be when it catches up to the first train?

Worked solution

1. Convert the second speed: $25 \text{ m/s} = 25 \times 3.6 = 90 \text{ km/h}$.
2. Let t be hours since the first train left. Equal distances: $60t = 90(t - 1.5)$.
3. $60t = 90t - 135 \Rightarrow 30t = 135 \Rightarrow t = 4.5$ hours.
4. Distance: $60 \times 4.5 = 270 \text{ km}$.

Answer: 270

27.

SAT CHALLENGE · CALCULATOR

Water flows through a cylindrical pipe with a circular cross section of radius 5 centimeters at a constant speed of 2 meters per second. To the nearest liter, how many liters of water pass through the pipe in one minute? (1 L = 1000 cm³.)

Worked solution

1. Cross-sectional area: $\pi r^2 = 25\pi$ cm². Speed: 2 m/s = 200 cm/s.
2. Flow rate: $25\pi \times 200 = 5000\pi$ cm³/s = 5π L/s.
3. Per minute: $5\pi \times 60 = 300\pi \approx 942.48$.
4. To the nearest liter: 942.

Answer: 942

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

Two solid statues are geometrically similar and made of the same uniform material. The smaller statue has a mass of 16 kilograms and a surface area of 0.8 square meters. The larger statue has a surface area of 5 square meters. What is the mass of the larger statue, in kilograms?

(A) 250 ✓

(B) 400

(C) 40

(D) 100

Worked solution

1. Area ratio: $\frac{5}{0.8} = 6.25 = k^2$, so the scale factor is $k = 2.5$.
2. Mass scales with volume, by $k^3 = 2.5^3 = 15.625$.
3. Mass: $16 \times 15.625 = 250$ kg. (Using 6.25 instead of k^3 gives 100.)

Answer: (A)

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

A car travels m miles per gallon of fuel, and fuel costs d dollars per gallon. Given that 1 mile = 1.6 kilometers, which of the following expressions gives the fuel cost, in dollars, of driving x kilometers?

(A) $\frac{xm}{1.6d}$ (B) $\frac{xd}{1.6m}$ ✓(C) $\frac{1.6x}{md}$ (D) $\frac{1.6xd}{m}$ **Worked solution**

1. Miles driven: $\frac{x}{1.6}$. Gallons used: $\frac{x}{1.6m}$.
2. Cost = $\frac{x}{1.6m} \cdot d = \frac{xd}{1.6m}$.
3. Check with $m = 30$, $d = 4$, $x = 240$: 150 miles, 5 gallons, \$20; and $\frac{240 \cdot 4}{1.6 \cdot 30} = 20$.

Answer: (B)

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

Cube A and Cube B are made of the same material. The edge length of Cube A is 8 centimeters, and the mass of Cube B is 72.8% greater than the mass of Cube A. What is the edge length of Cube B, in centimeters?

(A) 10.5

(B) 13.8

(C) 9.6 ✓

(D) 9.9

Worked solution

1. Mass ratio: $1 + 0.728 = 1.728$. Same material, so the mass ratio equals the volume ratio.

2. Edge ratio: $\sqrt[3]{1.728} = 1.2$ (since $1.2^3 = 1.728$).

3. Edge of B: $8 \times 1.2 = 9.6$ cm. (Adding $72.8\% \div 3$ to the edge gives 9.9; taking a square root gives 10.5.)

Answer: (C)